

Leveraging AI for enhancing emotional and cognitive well-being in elderly populations

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Abstract— As the global population ages, more than 50 million people suffer from dementia, and challenges related to loneliness, memory loss, and cognitive decline are becoming increasingly worrisome. One in four elderly people feel isolated, primarily because family members or close friends are unable to maintain frequent contact, especially when living far away. To address this, we propose *CompaniOn*, an AI-based virtual companion that provides personalized emotional support and cognitive stimulation tailored to elderly users. By leveraging natural language processing and sentiment analysis, the application fosters meaningful interaction, supports health monitoring, and strengthens family connections. Model validation was performed with 20 elderly participants over a 14-day period. Results showed an average reduction of 10.7 points on the UCLA Loneliness Scale, with 79% of users experiencing a significant reduction in loneliness ($p < 0.05$). These findings demonstrate that AI-driven companionship can improve emotional well-being, reduce isolation, and serve as a scalable digital health solution for elderly populations.

Keywords — *virtual companion, artificial intelligence, loneliness, cognitive decline, memory loss.*

I. INTRODUCTION

According to the World Health Organization (WHO), there were 1 billion people aged 60 years and older in 2020, and this number is projected to reach 2.1 billion by 2050 [1]. The rapid growth of the elderly population significantly increases the demand for solutions that assist with daily activities, monitor health—particularly mental health—and provide emotional support. Elderly individuals often live alone due to the death of a spouse or geographic separation from their children. As a result, they are highly exposed to social isolation, which is one of the most serious challenges associated with aging. Research shows that loneliness and isolation substantially increase the risk of premature mortality [2]. These findings highlight the urgent need for innovative approaches that can provide companionship and mental stimulation to elderly populations.

Although modern medicine continues to extend life expectancy, it also contributes to a higher prevalence of chronic conditions such as diabetes, cardiovascular illnesses, and neurodegenerative disorders including dementia and Alzheimer’s disease. These conditions further strain healthcare systems and increase the complexity of elderly care. At the same time, the global elderly care services market, valued at 1,025 billion USD in 2021, is projected to nearly double to 1,965 billion USD by 2032 [3]. Services such as assisted living, hospice, and institutional care remain essential but are costly, resource-intensive, and often unable to meet the emotional and social needs of older adults.

While digital solutions have emerged to address parts of this problem, they are typically fragmented. Applications such as

conversational AI chatbots provide emotional support, while cognitive training platforms focus on memory enhancement and brain exercises. Other services connect older adults with human companions to assist with everyday tasks. However, very few solutions offer an integrated approach that simultaneously addresses emotional companionship, cognitive stimulation, health monitoring, and family communication. This lack of a unified platform leaves many elderly individuals underserved and limits caregivers in their ability to provide holistic support.

To bridge this gap, this study introduces *CompaniOn*, an AI-based companion application designed to improve both the emotional and cognitive well-being of elderly users. The system aims to reduce loneliness through empathetic conversational interaction, foster cognitive engagement through reminders and dialogue, strengthen family bonds via seamless communication tools, and enhance safety by integrating real-time health monitoring, location tracking, and emergency alerts. In doing so, it provides a comprehensive solution that goes beyond the limitations of existing tools.

The main contributions of this work include the development of an integrated AI system tailored to elderly populations, empirical validation of its effectiveness in reducing loneliness through a user study, and the presentation of a hybrid architecture that combines cloud-based AI services, real-time monitoring, and a user-friendly interface. The remainder of this paper is organized as follows: Section II reviews related work and competing solutions in the field of elderly care technologies. Section III introduces the proposed model and describes its system architecture and design. Section IV presents the methodology and results of the model validation study. Section V concludes with a discussion of key findings and directions for future development.

II. STATE OF THE ART

Over the past decade, the rapid expansion of the elderly care market has led to the emergence of numerous digital tools designed to support older adults. These solutions address domains such as emotional companionship, cognitive training, and caregiving assistance. However, most remain limited in scope and fail to provide a comprehensive approach tailored to the combined emotional, cognitive, and safety-related needs of elderly users.

One prominent example is *Replika*, an AI chatbot that creates personalized conversations using natural language processing [4]. While *Replika* helps users feel emotionally connected, it is designed for a general audience and does not incorporate elderly-specific features such as medication reminders, fall detection, or family communication. This limits its usefulness in addressing the unique challenges faced by aging populations.

In contrast, *PapaPal* focuses on practical, real-world support by connecting elderly individuals with trained human companions [5]. These companions assist with daily tasks, transportation, and social interaction. Although effective in reducing isolation during in-person

sessions, the approach lacks digital scalability and does not integrate emergency alerts, remote monitoring, or AI-driven interaction.

Other platforms such as Lumosity and CogniFit target cognitive well-being by offering exercises and brain-training activities designed to enhance memory, attention, and problem-solving skills [6,7]. These applications are valuable for stimulating mental activity, but they do not provide emotional support, health monitoring, or mechanisms for maintaining family bonds.

Taken together, these solutions illustrate the fragmented nature of existing technologies for elderly care. Emotional support, cognitive training, and daily assistance are often addressed in isolation, leaving a gap for integrated systems that can combine multiple dimensions of support within one platform. Addressing this gap is the motivation behind the CompaniOn application, which seeks to merge emotional companionship, cognitive engagement, health and safety monitoring, and family connectivity into a single, user-friendly system.

CompaniOn integrates the most important services into one software solution:

- Real-time 3d avatar which acts both as friend and caregiver
- Reminders for therapy
- Real-time and previous locations tracking
- Health monitoring and emergency alert systems
- Communication with family members

III. PROPOSED MODEL

The application itself, even though it uses a third-party API as a chatbot functionality, consists as a rather complex mix of different knit up technologies that are as follows:

- Chatbot mechanism through a 3D animated model.
- .NET - A C# Framework used for creating the backend endpoint that oversees communicating with a traditional SQL based database and then bringing that information back to the UI in a presentable way to the user.
- User UI – An interface made in Flutter, designed to connect the 3D AI chatbot to the user together with the endpoints created in .NET. It is also important to note the fact that the application is focused on 2 user groups, mainly elderly users and their family members or caregivers, therefore the UI part of the application consists of 2 separate designs where only the elderly user have access to the chatbot mechanisms of the 3D animated model, while the family members have more of an overview design from which they can control and monitor the elderly user.
- Location Tracking: Continuously monitors and records the user's real-time geographic position to support caregivers and ensure safety, especially in cases of disorientation or wandering.
- SOS Notifications: Enables users to send immediate emergency alerts to predefined contacts or services, including their current location, with a single tap or voice command.

Since the chatbot mechanism represents a fundamental element of the proposed AI model and its practical application, the architectural design, implementation strategy, and functional dynamics will be presented in the remainder of the paper.

All components previously outlined, which are essential for the effective implementation of the proposed model addressing the

challenges introduced in the opening section, are depicted and further elaborated in Fig. 1.

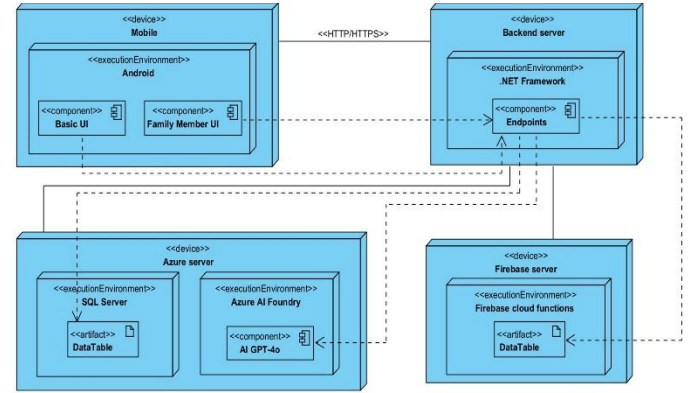


Fig. 1: The components of the CompaniOn.

The chatbot mechanism is implemented on the Basic UI side because it was opted to differentiate the Basic user interfaces from the Family member user interface. That is done so that the Basic user can focus on the chatbot interaction while the family member can focus on tracking the well-being of their elderly family members through their services.

As it was said before, the .NET Framework is being used to implement the endpoints that are being accessed from the user interfaces. Fig. 1 shows that the user interfaces are dependent on the endpoints.

External resources are used to integrate services into the application more smoothly; those resources are the Azure server and the Firebase server. The Azure server was used for two main purposes, firstly the hosting of the SQL database which is a main data storage location, and secondly for the access of Azure OpenAI models, in proposed model it is GPT-4o. Azure OpenAI lets users easily implement Content generation and analysis, summarization, information search and automation of certain tasks through their AI models. In short, with the implementation of a service like Azure OpenAI the application is modernized so it can use generative AI and even cloud operations if it would seem beneficiary [9]. The application also uses the Firebase server as a way of storage for important notifications that need to always be available to the different user groups. Firebase was chosen for its ability to act as a NOSQL database which comes in handy while handling notifications that need to be broadcast between users in a swift manner.

A. Frontend Chatbot Integration

Frontend Chatbot integration is presented in Fig. 2 while App UI is shown in Fig. 5. The flow starts when user send input to the 3D animated avatar model. The user has two basic ways of input. The first is classical input via text while the second one is input that refers to voice commands.

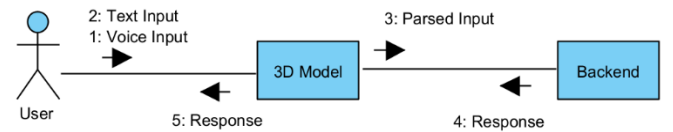


Fig. 2: Chatbot communication flow

For CompaniOn App the recommended input method are the voice commands as they enable users to speed up their input, the users can feel less overwhelmed, and voice commands help users with the forming of a trusty and emotional connection to the whole application [10]. To gather the voice input, a working microphone on the mobile device is needed. The application listens to the microphone input of the device, where it's set so that automatically if the input is nonexistent for a period longer than 5 seconds, the listening mechanism is stopped. Taking into consideration that the 3D model is animated, the whole time a listening animation or an idle animation will be played by the 3D model if the user is giving an input. At the same time as the user is giving the commands, the input is parsed to text via the Flutter Speech to Text Library [11]. The parsed input text is then displayed to the user, as he is giving out the voice commands. After a period of microphone silence longer than 5 seconds, or if the user manually turns off their microphone, the input is sent to the backend for processing.

After the information is processed on the backend and sent to the Azure AI services, the result needs to be displayed to the end user. Considering that the user expects to have an interactive conversation with the 3D animated model, the response is parsed by Flutter Text to Speech Library from a textual form to an audio form [12]. As soon as the parsing begins, a talking animation of the 3D model is played to simulate the audio coming from the 3D model. At the same time, captions that look like subtitles are displayed underneath the model so that the user has visual and audio feedback of the information that the chatbot is communicating.

When the process of the response recitation is finished, the 3D model goes back to either the idle animation or the listening animation, depending if the microphone is on or not. The sequence of gathering information and receiving information together with the playing of the animations of the 3D model represent a communication cycle for the elder user. The whole process can be repeated as many times as the user feels the need for a conversation.

B. Backend Chatbot Integration

The backend integration of the chatbot is done through a dedicated service that corresponds to the repository pattern that is followed by the whole project.

The first step after securing a connection to the Azure Open AI Service is to create a custom prompt with prompt engineering. This is done so that the possibility of getting an adequate and relevant answer to the user input is maximized, which is especially important in the field of medicine [13]. The prompt that is used is: "You are an AI assistant whose task is to talk to elderly people and be a friend to them. Keep your answers a bit short. Here is the question the user had:". At the end of the prompt, the user input is added, together with the previous question that the user asked and the chatbots response to that previous question. This is done so that the chatbot has more context, it is also a simple way of achieving a conversation history feature.

The LLM model that is used by Azure Open AI is GPT-4o, which is also the reason why such a large prompt, together with the conversation history can be forwarded. GPT-4o has a significant context window size and tokenization efficiency [14] which are used to generate a response that takes into consideration all the previous aspects and nuances of the conversations the user had with the model.

The final step of the backend process is to take the generated response for the user input, and to make a sentiment analysis. The sentiment analysis is done by instructing the LLM model to analyze its own response and the user input. If the user input or the response has any harmful comments, expressions of depression, loneliness, or other physical problems, the AI should flag that message as alarming. If a message is flagged as alarming, it will be forwarded to the family members as an urgent notification. If a message is not alarming, it will

be flagged as normal and the response to it will be normally displayed to the user and not sent to the family members.

IV. MODEL VALIDATION

Model validation is based on investigating whether AI conversations can affect the feeling of loneliness. For this investigation the data from 20 users was collected. Elderly people were using the app during the test period of 14 days. Users underwent the loneliness test before and after the test period using UCLA Loneliness Scale consisting of 20 questions. The UCLA Loneliness Scale is a commonly used measure of loneliness [15]. Questions measure different indicators of loneliness, and every question is marked with value 1-4 (From 1-Never to 4-often). Before the use of the app, users went through questions from the scale. Examples of the questions: "How often do you feel that you lack companionship?", "How often do you feel like there is no one to turn to?", "How often do you feel alone". After completing the initial questionnaire, and collecting scale results, the users engaged in a 14-day trial period using the test version of the AI application. Upon completion of this period, they filled out the same questionnaire again in order to assess potential changes in their sense of loneliness.

A. Statistical Indicators

In the process of validation of the model, analysis of statistical indicators was done with the goal to access efficiency of the app with reducing the level of loneliness of the elderly users. The analysis included data collected from 20 users, where loneliness levels were measured before and after using the application. Statistical indicators include median, minimum, maximum and range analysis.

Average loneliness measured before the test period was 61.05, while that value dropped by 10.7 points on the UCLA scale at the end of the test period. After the use of the CompaniOn App average loneliness is 50.8.

Minimum and maximum: The largest decreases in loneliness levels were observed in users 13 (a decrease of 12 points) and 17 (a decrease of 14 points), while the smallest decreases were in users 3 and 10, who reduced their loneliness levels by 3 points.

Standard deviation: The standard deviation of the measured values was 4.25, suggesting that even there is significant improvement in most users, there is also some degree of variability in response to the app among different users. Figure 3 presents a box plot illustrating the distribution of loneliness levels before and after using the app, with a noticeable reduction in median values and overall range, indicating a potential positive impact of the app on reducing loneliness. On the X-axis, two groups are presented: "Before using the app" and "After using the app." The Y-axis represents numerical values indicating the level of loneliness, ranging approximately from 40 to 70.

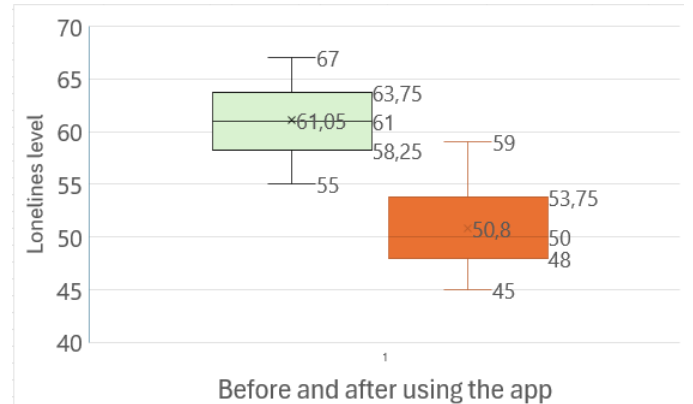


Fig. 3: UCLA Loneliness Scores Pre and Post interaction with AI

B. Model Accuracy

Accuracy of the model is evaluated through analyzing the responses of the CompaniOn app regarding the reduction of loneliness. A model is considered accurate if the loneliness of the user is reduced by a significant amount in short period of time. For the period of 14 days significant amount was considered reduction of loneliness greater than 5 points on the UCLA scale. Accuracy analysis included all the users and compared changes in the level of loneliness before and after using the application.

The overall accuracy of the model in reducing loneliness was 79%, which means that 79% of users experienced a significant reduction in loneliness.

Accuracy was higher for users with higher initial levels of loneliness. Users which before using the application had a higher level of loneliness (over 60 points) achieved a greater reduction in the level of loneliness.

Given that the accuracy rate is far above the reference 50%, it can be conclude that the system has relatively high accuracy in reducing the feeling of loneliness. Fig. 4 shows the user results for the decrease in loneliness where 79% of users had loneliness reduction greater than 5.

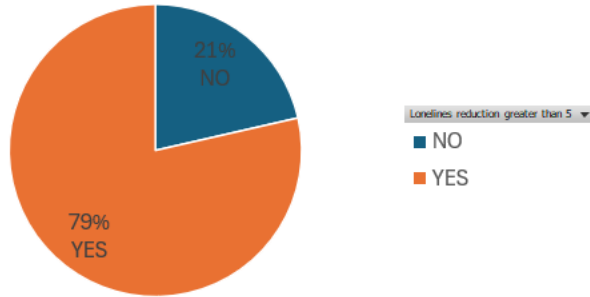


Fig. 4: User Results: Decrease in Loneliness

C. Hypothesis Testing

For testing the hypothesis paired t-test is used. A t-test is a type of statistical test that is used to compare the means of two groups. It is one of the most widely used statistical hypothesis tests in pain studies [16]. T-test is used to determine whether there is a significant statistical difference between the levels of loneliness before and after the use of the CompaniOn App. The null hypothesis (H_0) stated there is no significant difference in the level of loneliness before and after using the application, while the alternative hypothesis (H_1) stated: Using the application reduces loneliness of the elderly users.

Hypothesis testing: Using the t-test for paired samples (paired t-test), a t-value of 12.18 was calculated, and the p-value was 0.0001. This p-value is lower than the standard level of significance ($\alpha = 0.05$), which allows rejection of the null hypothesis.



Fig. 5: CompaniOn App UI

Based on the results of the t-test, it can be confirmed that there is a statistically significant difference in the level of loneliness before and after using the application. This result supports the alternative hypothesis that using the app reduces loneliness of older users.

V. CONCLUSION

Based on the given analysis it can be concluded that proposed AI model used within CompaniOn App represent innovative and practical solution for elderly users. Through this research is shown that AI can also be a useful tool for reducing level of loneliness. AI model has shown that it can be polite, compassionate and reliable companion. Result of the model validation shows that the application has a positive impact on the emotional state of elder users. The drop in UCLA loneliness scale is 10.4 points, while the accuracy of the model in reducing the feeling of loneliness is high (80%). Statistical analysis, including standard deviation and the application of the t-test for dependent samples, shows the existence of a significant difference between the level of loneliness before and after using the application ($p < 0.05$). This confirms the alternative hypothesis that the application contributes to the reduction of loneliness of the elderly users. With all the details it can be confirmed that AI can be powerful tool in loneliness reduction and building emotional relationships with users.

In addition to the functionality of the chatbot, additional features such as the SOS button, location tracking and notification system contribute to the competitive advantage of the CompaniOn App. The combination of traditional and modern technological solutions enables flexibility, security and scalability of the system.

It can be concluded that CompaniOn App represents valuable contribution in the field of digital health support for the elderly population, with the potential for further development and wider use of the application and features in the future.

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